DATA SCIENCE PROJECT

Cookie Cats

A/B Testing Analysis

Gate 30 vs Gate 40 · Retention · Bayesian Inference · ML Prediction

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What is A/B Testing?

A/B Testing is a controlled experiment where two variants (A = Control, B = Treatment) are shown to randomly split audiences. Statistical analysis determines which version performs better on a key metric.

① **Randomization**

Users are randomly assigned to groups — eliminating selection bias and ensuring a fair comparison between variants.

② **Hypothesis**

Define H_0 (no difference) and H_1 (variant differs). Reject or fail to reject H_0 based on the p-value threshold.

③ **Significance**

$p < 0.05$ means less than 5% chance the result is due to random chance. Not the same as business impact.

④ **Effect Size**

Cohen's h or relative lift % measures practical importance. A tiny but significant effect may not be worth acting on.

Project Overview: Cookie Cats

🎮 The Game

- Cookie Cats is one of the world's most downloaded mobile puzzle games.
- Players progress through levels interrupted by gates — forced stops requiring a wait or in-app purchase.
- The experiment tested moving gate from Level 30 → Level 40.
- Core question: Does gate position affect long-term player retention?

Group Distribution

📊 Variables

userid
Unique player ID

version
gate_30 / gate_40

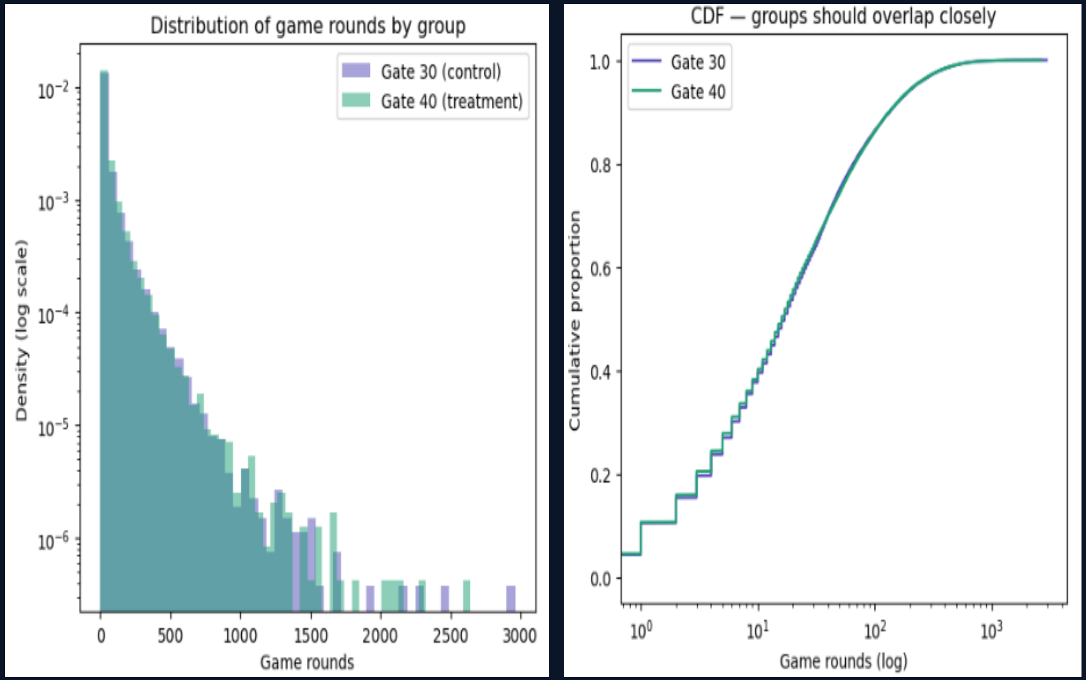
sum_gamerounds
Rounds in 14 days

retention_1
Return on Day 1?

retention_7
Return on Day 7?

Total Players: 90,189 | Gate 30 (Control): 49.6 % | Gate 40 (Treatment): 50.4 % | Observation: 14 Days

Data Cleaning & Exploratory Analysis



01

No Missing Values / No Duplicates

0 missing values, 0 duplicates. All userids unique → clean randomization confirmed.

02

Outlier Removed

User 6390605 played 49,854 rounds (46× p99). Bot behavior. Removed via log-IQR.

03

Right-Skewed Distribution

90% of players ≤ 134 rounds. Classic power-law. Log transform used for analysis.

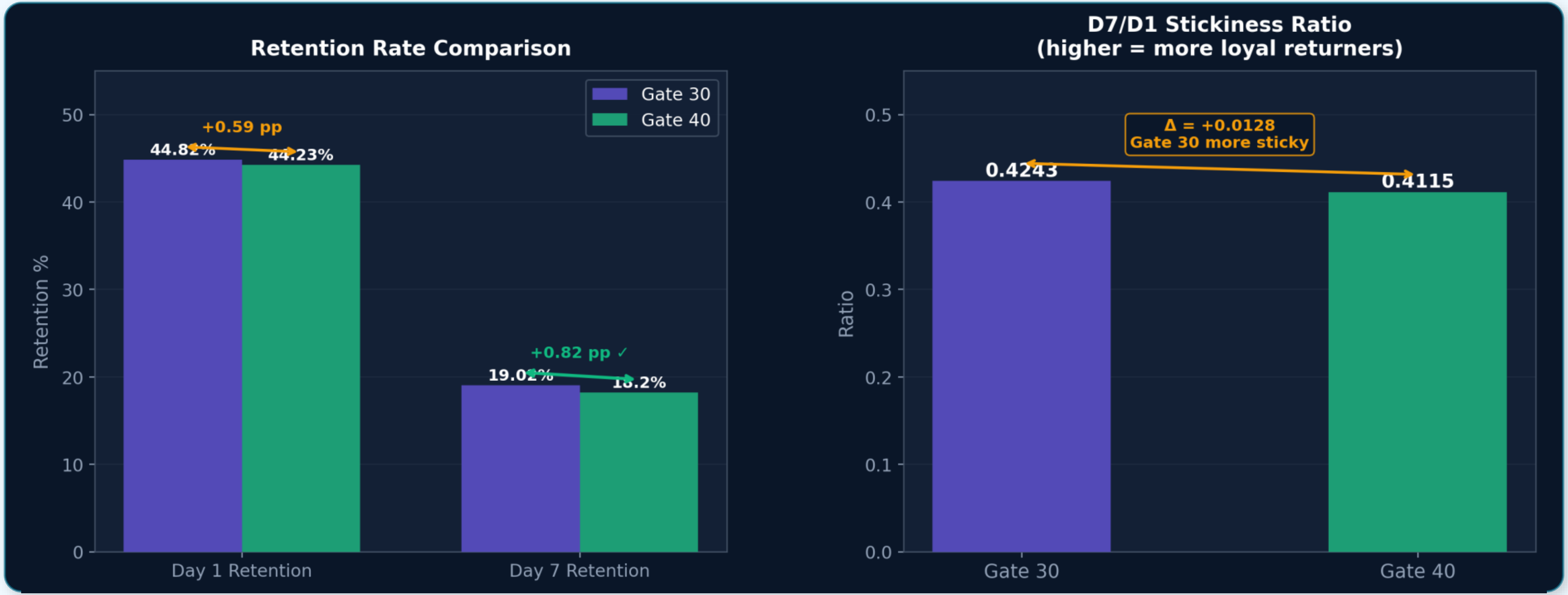
04

Groups Are Balanced

CDF and histogram overlays confirm near-identical distributions between Gate 30 and Gate 40 — validating randomization. Median: ~17 rounds (Gate 30) vs ~16 rounds (Gate 40).

💡 Insight: Heavily right-skewed data. Non-parametric tests (Mann-Whitney U) are more appropriate than t-tests here.

Retention Results: Gate 30 vs Gate 40

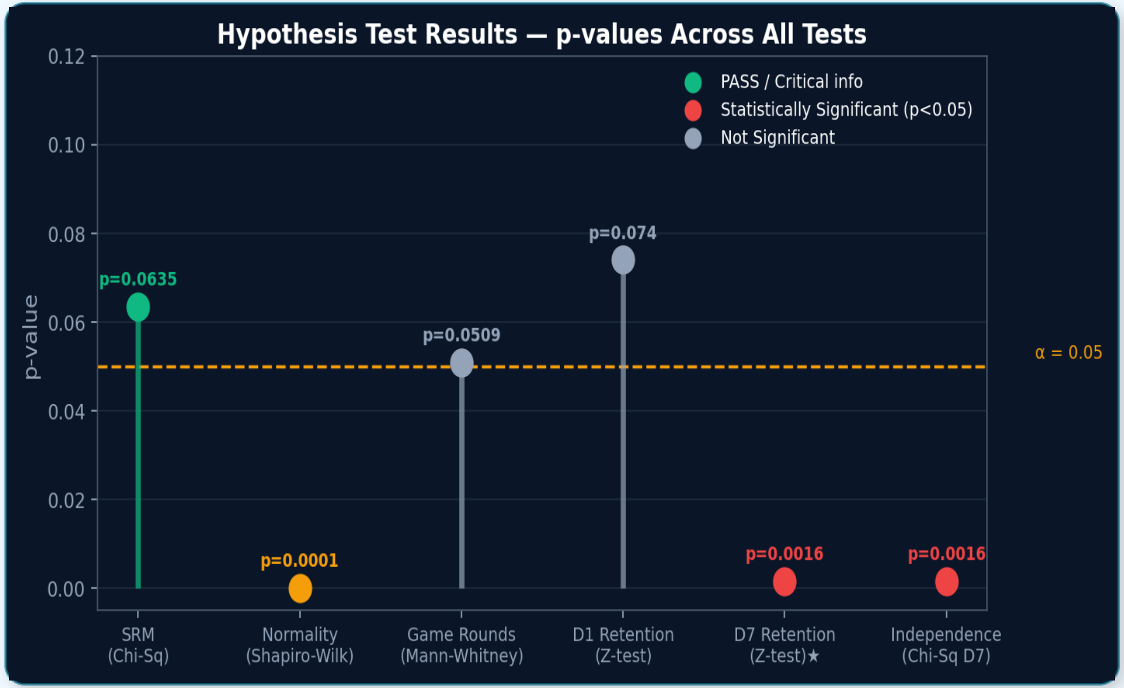


📅 Day 1: +0.59 pp | Not Significant
(pp : Percentage Points)

📅 Day 7 ★: +0.82 pp | p = 0.0016 ✓
(pp : Percentage Points)

📌 Key Insight: Gap grows | Habit Formation

Statistical Hypothesis Testing



SRM Test	**Randomization was valid. 50/50 split confirmed. A/B test results are reliable.**
Shapiro-Wilk Normality Test	**Game rounds are NOT normally distributed (right-skewed). Non-parametric tests selected.**
Mann-Whitney U Test	**No significant difference in total rounds played. Gate position doesn't change gameplay volume.**
2 Proportion Z- Test (D1)	**Day 1 retention difference is not statistically significant. Gate doesn't affect immediate behavior.**
2 Proportion Z- Test (D7)	**Day 7 retention IS significantly higher for Gate 30. Strong evidence of long-term habit formation.**
Chi-Sq Independence Test (D7)	**Confirms Z-test: D7 retention depends on gate version. Gate placement affects long-term behavior.**

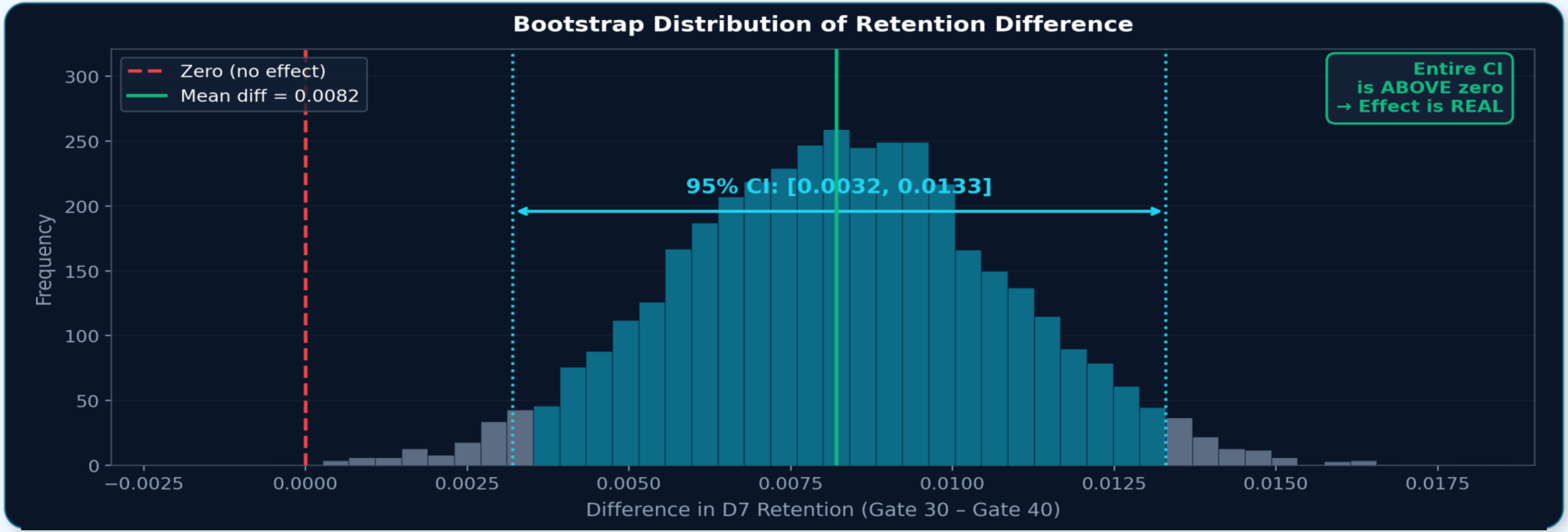
💡 Only D7 retention is statistically significant. Day 1 is not — Gate 30's advantage compounds over time rather than showing immediately.

Effect Size · Power Analysis · Bootstrap CI

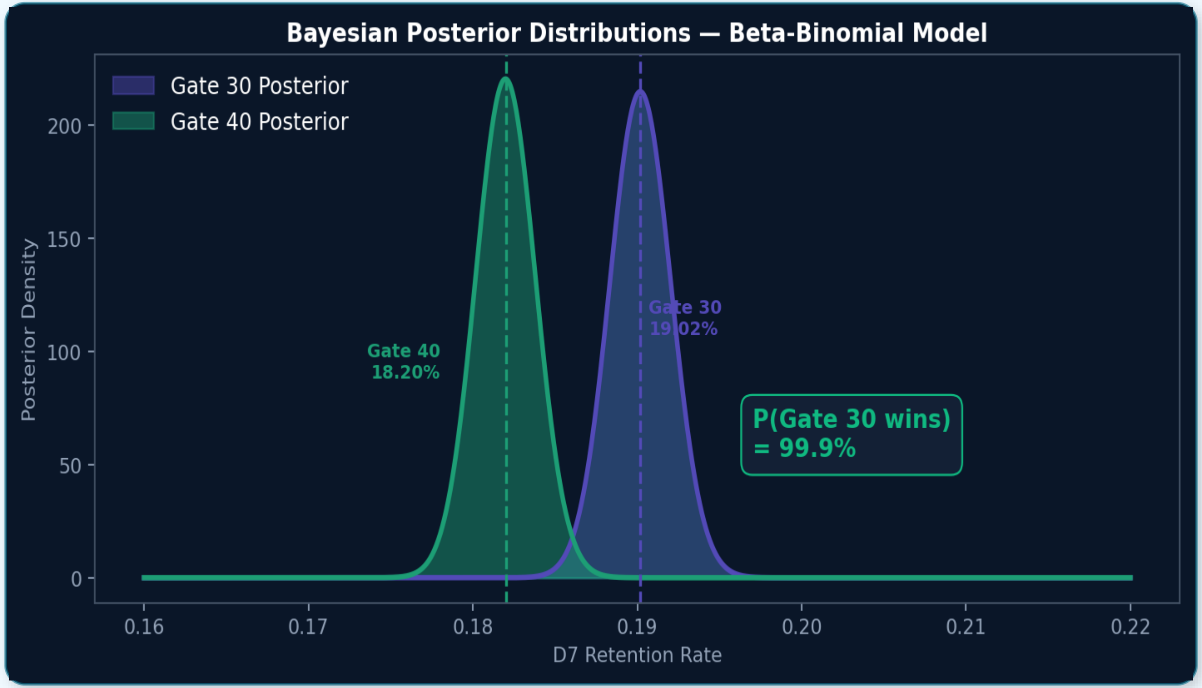
0.021
Cohen's h Effect Size
Small (< 0.2 threshold)

44,699
Actual Sample / Group
Required: 35,502 → Powered ✓

99.9%
P(Gate 30 > Gate 40)
Bootstrap over 10000 simulations



Bayesian A/B Testing



How It Works

1. Prior: Beta(1,1) — no assumption
2. Observe successes & failures
3. Posterior: Beta(1+s, 1+f)
4. Sample 100K draws per group
5. P(30 wins) = fraction where 30 > 40

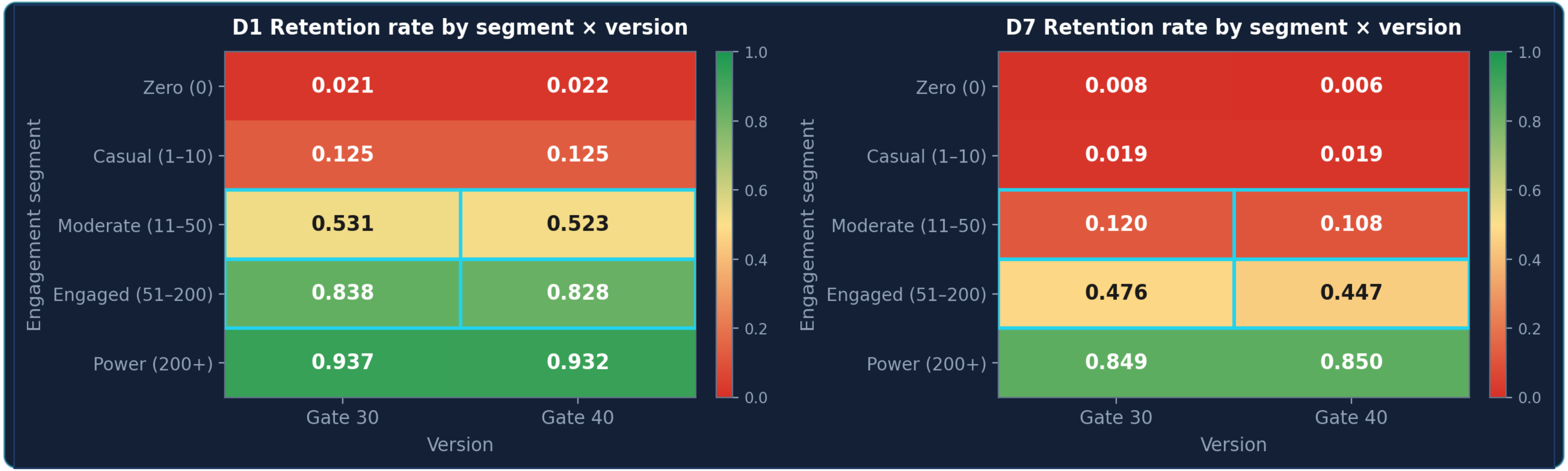
99.9%

P(Gate 30 wins)

Beta-Binomial Posterior

💬 Business Language: "There is a 99.9% probability Gate 30 produces higher 7-day retention than Gate 40." · Est. Revenue Impact: ~ $42,795/yr

Segmentation Analysis: Retention by Engagement Segment



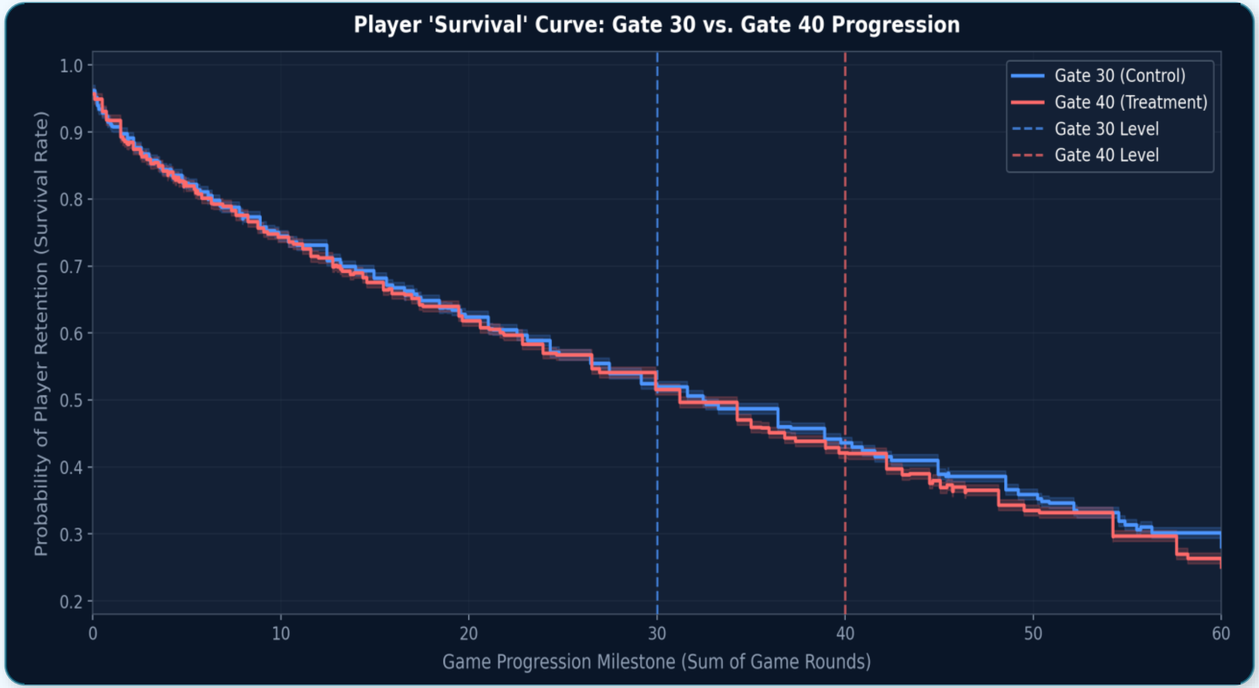
Segment	n (Gate 30)	n (Gate 40)	Diff of D7 (pp)	p-value	Sig?
Zero (0)	1,937	2,057	+0.19	0.4704	Not sig
Casual (1–10)	15,736	16,259	-0.01	0.9715	Not sig
Moderate (11–50)	15,891	15,497	+1.21	0.0007	✓ Sig
Engaged (51–200)	8,531	9,058	+2.88	0.0001	✓ Sig
Power (200+)	2,604	2,618	-0.12	0.9043	Not sig

Casual players (1–10 rounds): NO significant difference — gate not yet encountered

Moderate (11–50) & Engaged (51–200): SIGNIFICANT — these players decide whether to stay

Power users (200+ rounds): No sig. difference — they adapt regardless of gate position

Survival Analysis: Player Retention Journey



Log-Rank Test

$p = 0.023 < 0.05$ → Curves significantly different

Before Level 30

Both curves identical → randomization confirmed. No pre-treatment bias.

Levels 30–40

Visible divergence: Gate 30 retains better as players hit the gate and continue.

After Level 40

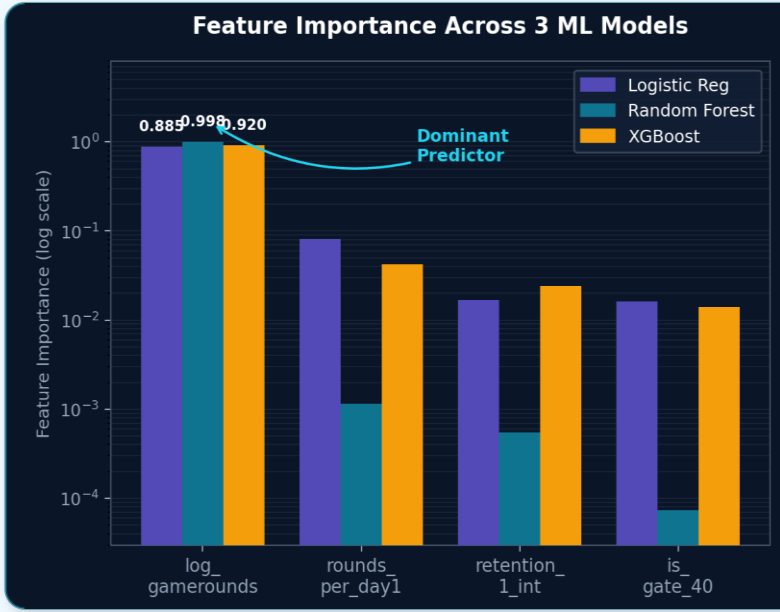
Gate 30 survival curve stays elevated. The advantage persists throughout.

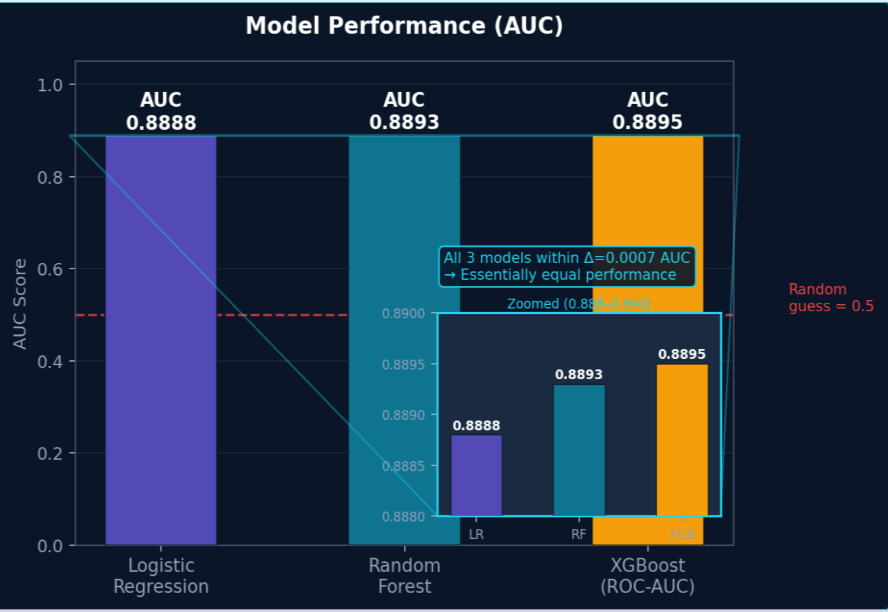
Business Meaning

Gate position doesn't just change one metric — it changes the entire player journey.

💡 Kaplan-Meier reveals the full churn story across time — not just a single retention snapshot.

Predictive Modelling: What Drives Retention?





🏆 Engagement Wins

log_gamerounds is the #1 predictor across all 3 models — far ahead of gate version.

🔬 Gate Matters, But...

is_gate_40 ranks last. Gate design influences engagement — but engagement predicts retention.

💡 Product Implication

To improve D7 retention: focus on getting players to play more rounds in the first 14 days.

What I'd Do Next — Production Roadmap

This analysis answers the business question. These are the next steps to make the experiment framework production-grade.

01

Variance Reduction

CUPED — Controlled-experiment Using Pre-Experiment Data

Use pre-experiment engagement data as a covariate to reduce outcome variance. Tighter variance → smaller required sample → faster, more sensitive experiments.

Why it matters for this project:

Would have tightened our CI from [0.003, 0.013] — detecting the effect with fewer players.

02

Metric Design

Define a Composite OEC (Overall Evaluation Criterion)

D7 retention alone doesn't capture revenue, session length, or monetisation. A weighted OEC prevents optimising one metric at the expense of others.

Why it matters for this project:

Gate 30 wins on retention — but does it affect in-app purchase rate? An OEC would answer this.

03

Experiment Design

Sequential Testing & Alpha Spending

Sequential testing (e.g. Lan-DeMets alpha spending) allows safe early stopping without inflating the false positive rate — unlike standard fixed-horizon tests.

Why it matters for this project:

Would let the product team ship Gate 30 faster if significance is reached early, reducing opportunity cost.

04

Validity Check

Network Effects & Cluster Randomisation Check

If players in the same social cluster are split across groups, SUTVA is violated — treatment leaks. Check for interference before trusting individual-level randomisation.

Why it matters for this project:

Cookie Cats has social features — cluster-level analysis would confirm whether peer effects contaminate results.

05

Longitudinal Validation

30-Day Holdback & Long-Window Retention

D7 captures short-term habit. A 30-day holdback cohort (small group kept on Gate 40) confirms whether Gate 30's advantage persists or narrows over time.

Why it matters for this project:

Some gate effects are novelty-driven and fade. A holdback confirms the result is durable, not temporary.

Key Takeaways & Technical Learnings

📌 Statistical Significance ≠ Business Significance

Cohen's h = 0.021 is statistically significant due to the large sample (90K users) but practically small. Always report effect size — not just p-values — in business decisions.

⚠️ Always Check SRM Before Anything

If randomization fails, every downstream result is invalid — regardless of the p-value. SRM check (χ^2 p = 0.0635 → PASS) was the first step here, not an afterthought.

📌 Segmentation Reveals the True Drivers

Overall results can be misleading. The real Gate 30 benefit lives in Moderate and Engaged players (11–200 rounds). Blanket recommendations miss this nuance.

⚠️ Large Samples Detect Even Tiny Effects

With 90K users, even a 0.001 difference becomes significant. This is why effect size must always accompany p-values — significance alone misleads at scale.

📌 ML Explains What Statistics Confirm

XGBoost + SHAP revealed that engagement features dominate prediction. ML is not a replacement for A/B testing — it complements it by building a churn prediction model.

⚠️ Bayesian Probability Speaks Business Language

"p = 0.0016" confuses stakeholders. "99.9% probability Gate 30 is better" closes rooms. Lead with posterior probability in presentations — keep p-values in the appendix.

Thank You

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